

Trends and Cycles II

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1 Recap

What we now know

- For estimation, we prefer working with stationary data and models
- Most economic data is non-stationary
- We can remove the non-stationary part using different approaches
 - Differencing the data, i.e., looking at the (first) derivatives / growth
 - * But then miss the level which might end up important
 - Subtract trend component, but we aware of spurious cycles
 - * Deterministic trend is easy, but is it realistic?
 - * Stochastic trends

But often in economics, we are interested in the trend/cycle per se. Not because of the statistical issues involved with non-stationary data, but because theory is written in terms of trends or cycles.

2 Trends and cycles

In the following we will assume that (the log of) a time series can be decomposed into two components, a trend (g_t) and a cycle (c_t):

$$y_t = g_t + c_t$$

To obtain estimates of (g_t) and (c_t), we will consider several univariate detrending methods.

3 Deterministic trend

The traditional method used to estimate business cycles was to define a smooth (natural) growth path for the economy, which was only perturbed by transitory cyclical fluctuations

$$\begin{aligned}y_t &= g_t + c_t \\ \hat{g}_t &= \hat{\alpha}_0 + \hat{\alpha}_1 t + \hat{\alpha}_2 t^2 + \dots \\ \hat{c}_t &= y_t - \hat{g}_t\end{aligned}$$

The assumption of a linear trend (LT) implies $|\alpha_1| > 0$, and $\alpha_2 = 0$.

For a quadratic trend (QT), one assumes $|\alpha_1| > 0$, and $\alpha_2 > 0$.

4 Linear filters

A filter is a function that transforms a time series into another time series. If we let g_t represent a moving average of observed y_t , we can extract the trend component (g_t) and the cyclical component (c_t). That is, let $g_t = G(L)y_t$, or:

$$g_t = \sum_{j=-m}^n g_j y_{t-j} \quad (1)$$

where $m, n \in \mathbb{Z}^+$ and g_j are the weights in the $G(L)$ polynomial:

$$G(L) = \sum_{j=-m}^n g_j L^j \quad (2)$$

where L is defined so that $L^j y_t = y_{t-j}$ for positive and negative values of j . We will mostly focus our attention to symmetric moving averages, i.e., those for which weights are s.t. $g_j = g_{-j}$.

Equations (1) and (2) define the trend component. The cyclical component is simply the difference between the observed value of y and this component, i.e.:

$$c_t = [1 - G(L)]y_t \equiv C(L)y_t$$

We call $C(L)$ and $G(L)$ linear filters. Often the weights are chosen to add up to one, i.e., $\sum_{j=-m}^n c_j = 1$, so that the level of the series is maintained.

The 5-point moving average filter displayed in lec 1 is an example of a filter that was used to smooth the series. The weight was $1/5$, which was applied to a moving window of five observations. That way we created a cyclical component by filtering the noise, i.e.,

$$c_t = 1/5 \sum_{j=-2}^2 \epsilon_{t-j} = 1/5(\epsilon_{t-2} + \epsilon_{t-1} + \epsilon_t + \epsilon_{t+1} + \epsilon_{t+2})$$

Note how applying this type of filters to the data changes its properties!

One implication: two series that are totally unrelated can become highly related after applying a trend/cycle decomposition

5 The Hodrick-Prescott (HP) filter

A popular way to extract business cycles. The filter extracts a stochastic trend, (g_t) , which for a given value of λ , moves smoothly over time and is uncorrelated with the cycle. The ‘trend’ can be obtained as the solution to the problem:

$$\min_{[g_t]} \sum_{t=1}^T [(y_t - g_t)^2 + \lambda[(g_{t+1} - g_t) - (g_t - g_{t-1})]^2]$$

λ is the smoothing parameter. It “penalizes” the acceleration in the growth component.

- If λ approaches infinity, the lowest minimum is achieved when the variability in the trend is zero, and the trend is perfectly log linear.
- A small value of λ will allow more variation in the trend. If $\lambda = 0$, there will be no cycle, as the trend will follow the observed series perfectly.

Warning!

The HP filter is often used in economics. Use it with care:

- HP filter produces series with spurious dynamic relations that have no basis in the underlying data-generating process.
- The end-of-sample problem: HP filter produces cyclical series that are close to the observed data at the beginning and at the end of the estimation period.
- How to set λ ? And what does it basically imply. $\lambda = 1600$ is often used for quarterly macro data, but when estimating λ (which can be done) one often get very different values
- Read Hamilton on why it sucks:

“(a) The Hodrick-Prescott (HP) filter introduces spurious dynamic relations that have no basis in the underlying data-generating process. (b) Filtered values at the end of the sample are very different from those in the middle and are also characterized by spurious dynamics. (c) A statistical formalization of the problem typically produces values for the smoothing parameter vastly at odds with common practice. (d) There is a better alternative. A regression of the variable at date t on the four most recent values as of date $t - h$ achieves all the objectives sought by users of the HP filter with none of its drawbacks.”

6 Beveridge-Nelson (BN) decomposition

Definition: The implied trend for a non-stationary time series can be defined as the minimum MSE forecast of the long-run level of the series (minus any drift)

An equivalent formulation is that BN trend is the present level of the series plus the infinite sum of the minimum MSE h -period ahead first difference forecasts.

Let y_t be integrated of first order, so that its first differences, Δy_t , are stationary and can be accurately forecasted using a stationary, e.g., AR(1) model:

$$(\Delta y_t - \alpha) = \phi(\Delta y_{t-1} - \alpha) + \epsilon_t$$

The minimum squared error (MSE) h -period ahead forecast of the first difference AR is (tbc):

$$\mathbb{E}[\Delta y_{t+h} - \alpha] = \phi^h(\Delta y_t - \alpha) \quad (3)$$

As by definition, the BN trend is the present level of the series plus the infinite sum of the minimum MSE h -period first difference forecasts, we have:

$$g_t = y_t + \lim_{H \rightarrow \infty} \sum_{h=1}^H \mathbb{E}[\Delta y_{t+h} - \alpha] \quad (4)$$

Substituting the expression for the h -period ahead forecast from equation (3) into the definition in (4), we get:

$$\begin{aligned} g_t &= y_t + \lim_{H \rightarrow \infty} \sum_{h=1}^H \phi^h(\Delta y_t - \alpha) \\ &= y_t + \frac{\phi}{1 - \phi}(\Delta y_t - \alpha) \end{aligned}$$

The BN cycle can easily be derived from the result above using the definition: $y_t = g_t + c_t$. Thus:

$$c_t = \frac{\phi}{1 - \phi}(\Delta y_t - \alpha)$$

6.1 Evaluating the BN filter

- The advantage of the BN method is that it is an appropriate method to extract cycles when a series is difference-stationary and hence using a linear model may infer spurious cycles.
- It allows the series to contain a unit root that can be highly volatile
- One disadvantage is that it is rather time-consuming, as one has to choose between different ARMA models that may give quite different results.
- Cochrane (1988) pointed out that although different ARMA models may fit the short-run properties of the first differences of an observed series, the forecast functions from these models may differ substantially. Since the trend-cycle decomposition in the BN method relies on the forecast properties of the ARMA models, these models may give very different trend-cycle decompositions.

7 Hamilton (2018) filter

Builds on similar idea as the BN decomposition, but perhaps even simpler. Builds on fixed horizon direct forecasting equation rather than iterative infinity approximation as in BN.

If y_t is integrated of some order d , do the regression

$$y_{t+h} = \alpha + \sum_{j=1}^P \beta_j y_{t-j+1} + v_{t+h}$$

with $p > d$. Then, as shown in Hamilton (2018), the residuals $\hat{v}_{t+h}$ will be stationary and approximate will the cyclical component for a broad class of underlying processes

Thus, $c_{t+g} \equiv \hat{v}_{t+h} = y_{t+h} - q_t$, from which we can easily obtain the trend as well.

7.1 Intuition

Assume y_t is $I(d=1)$, s.t. $y_t = y_{t-1} + u_t$, and $p = 1$ and $h = 2$ for simplicity, then the regression

$$y_{t+2} = \beta y_t + v_{t+2}$$

implies that $\hat{\beta} = 1$ and $\hat{v}_{t+2} = u_{t+2} + u_{t+1}$ because the forecast of an RW, process h periods into the future

- Is in expectation $\mathbb{E}[y_{t+h}|y_t] = y_t$. This gives $\hat{\beta} = 1$ above
- But has errors $y_{t+h} - \mathbb{E}[y_{t+h}|y_t] = \sum_{j=1}^h u_{t+j}$. In turn, this implies that $\hat{v}_{t+2} = u_{t+1} + u_{t+2}$ above

Clearly, since u_t is Gaussian white noise and stationary, v_{t+2} is stationary as well. Put differently, the sum of stationary components is stationary.

Note, if the RW assumption is correct, simply computing $y_t - y_{t-h} = \Delta^h y_t = (1 - L^h) y_t$ would give the “same” as OLS regression

Key benefits (relative to e.g., HP filter)

- As BN decomposition; very data driven
- Less likely that any inferred correlation between two filtered variables are due to the filtering method per se. Rather, a property of the data.
- Relative to BN, potentially more control, because h can be chosen by the researcher

Implementation:

- How to choose h : up to you. But, for macroeconomic data and business cycles, Hamilton recommends $h = 8$ for quarterly data, i.e., 2-year horizon forecast
- How to choose p : needs to be bigger than d (which is unknown). Hamilton recommends $p = 4$

8 Spurious cycles

Nelson and Kang (1981): If one detrend data that are generated by an RW using a smooth deterministic trend, one can infer spurious cycles in the data. The deterministic trend would over-estimate the persistence of the cycle, whereas the trend component would be under-estimated.

Yule-Slutsky effect: A cycle could be spurious, simply because it is a function of the filters applied to extract the cycle from economic data. I.e., using a summation procedure to smooth a series in addition to differentiating to remove the long-run trend, would generate a spurious cyclical pattern.

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